

Study details

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Abstract

Phosphatidylethanolamine (PE) is a ubiquitous lipid species in higher eukaryotes which resides preferentially in mitochondria. Here, we synthesized a multifunctionalized PE derivative (1) de-signed to identify PE-binding proteins in intact cells through photo-crosslinking and subsequent isolation and proteomic analysis of the PE-protein conjugates. Due to its aromatic caging group, 1 is initially concentrating in perinuclear membranes. After uncaging, rapid translocation to mitochondria was observed. Hence, the tool is useful for tracking PE location and for determining the PE interactome. A trifunctional phosphatidic acid (PA) derivative in compari-son was rapidly metabolized and is hence more limited in its use.

Lipid probes utilized

Trifunctional Phosphatidic Acid

Trifunctional Phosphatidylethanolamine

Cell line analyzed

HeLa

Species: *Homo sapiens*

Cell type: Epithelial (immortalized cervical cancer line)

Uncaging & Crosslinking timeline

Lipid Probe	Uptake time	Uncaging time	Interaction time	Crosslinking time
PA	60 min	5 min	15 min	5 min
PE	60 min	5 min	15 min	5 min

Lipid Probe	Uptake time	Uncaging time	Interaction time	Crosslinking time
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Mass spectrometry quantification method

16-channel Tandem Mass Tagging (TMT16)

ProteomeXchange details

ProteomeXchange title: Trifunctional lipid derivatives: PEâ€™s mi-tochondrial interactome

ProteomeXchange accession: PXD059960

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Project Webpage: <http://www.ebi.ac.uk/pride/archive/projects/PXD059960>

FTP Download: <https://ftp.pride.ebi.ac.uk/pride/data/archive/2025/01/PXD059960>

Additional sample preparation ?

Cell lysates were separated into membrane and cytosolic fractions. This publication reports the results of just the membrane fraction.

Data wrangling

Use the download button below to download the R script used to wrangle the authors' original submission into the data visualization tools on the Lipid Interactome:

Use the download buttons below to download the original data file prior to wrangling: